



AS Chemistry

21 Carbonyl compounds

Candidates should be able to:

- ☐ Describe the general structure and physical properties of carbonyls.
- ☐ Describe the reactions by which carbonyls can be produced.
- ☐ Describe the reactions of carbonyls including reduction and nucleophilic addition.
- ☐ Describe the test for carbonyl compounds (aldehydes and ketones).

Definitions

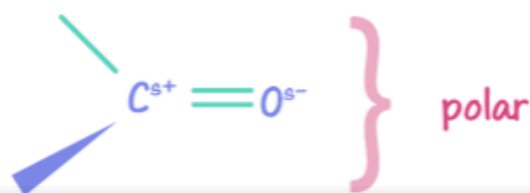
Nucleophilic addition	Formation of a carbon-oxygen single bond, when a nucleophile attacks an electron-deficient carbon atom of the unsaturated C=O bond.
Nucleophile	A negative or slightly negative species, which has at least one lone pair (electron donor).
Nitrile	A compound with the functional group -C≡N.

21.1 Aldehydes and Ketones

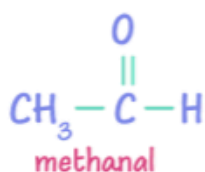
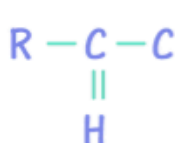
21.1.1 General structure and physical properties

Carbonyl group: carbon-oxygen double bond

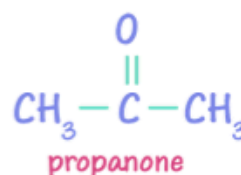
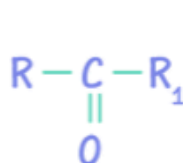
- Carbonyl group: carbon-oxygen double bond



- aldehydes



- ketones

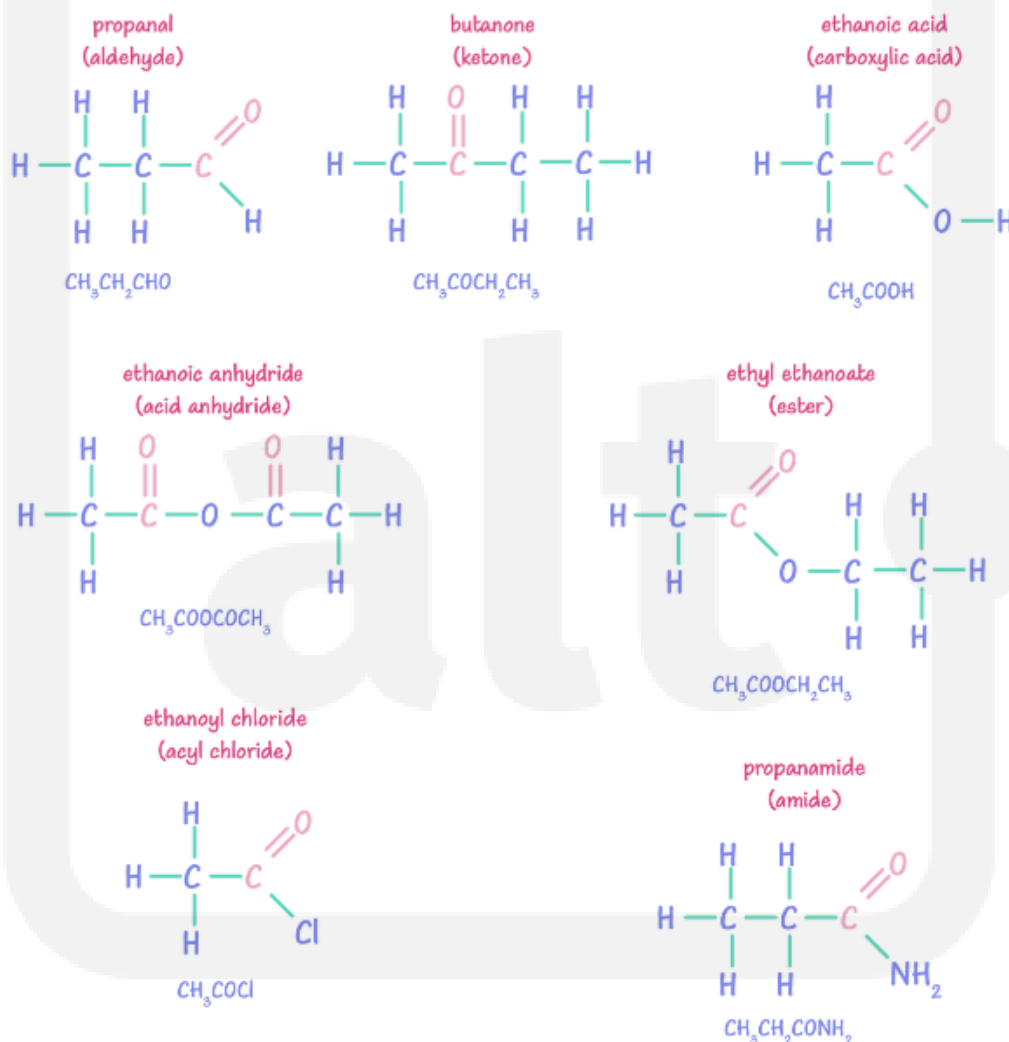




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- Aldehydes are easily oxidised because of the presence of Hydrogen atom, but ketones are not easily oxidised.
- sp^2 hybridisation is present
- planar, 120° compounds
- $[C=O]$ is present.
- aldehydes and ketones are functional isomers (CHO)
- permanent dipole-dipole attractions (intermolecular forces) are present.
- Hence, compounds with similar M_r have different boiling points.
- aldehydes and ketones can be considered as positional isomers of each other





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Name	Structural formula	Displayed formula
methanal	HCHO	
ethanal	CH ₃ CHO	
propanal	CH ₃ CH ₂ CHO	
butanal	CH ₃ CH ₂ CH ₂ CHO	

Name	Structural formula	Displayed formula
propanone	CH ₃ COCH ₃	
butanone	CH ₃ COCH ₂ CH ₃	

name	structural formula	displayed formula
methanal	HCHO	
ethanal	CH ₃ CHO	
propanal	CH ₃ CH ₂ CHO	
butanal	CH ₃ CH ₂ CH ₂ CHO	

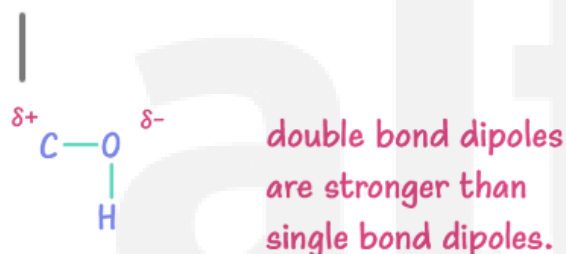
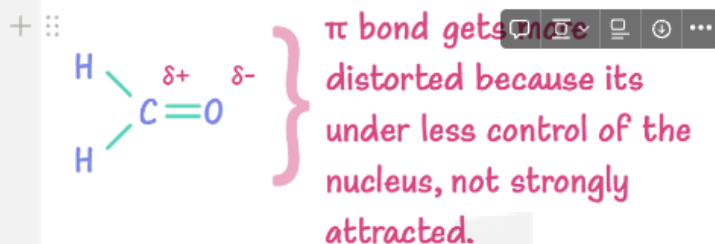
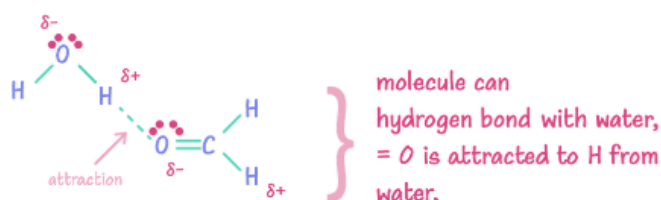
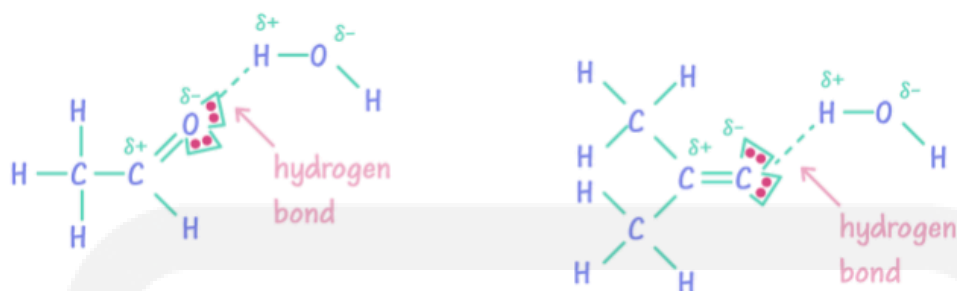
name	structural formula	displayed formula
propanone	CH ₃ COCH ₃	
butanone	CH ₃ COCH ₂ CH ₃	

- solubility: small molecules are soluble due to polarity, but larger ones won't be soluble, because non-polar portion of the molecule will be more



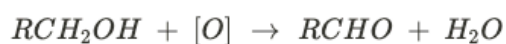
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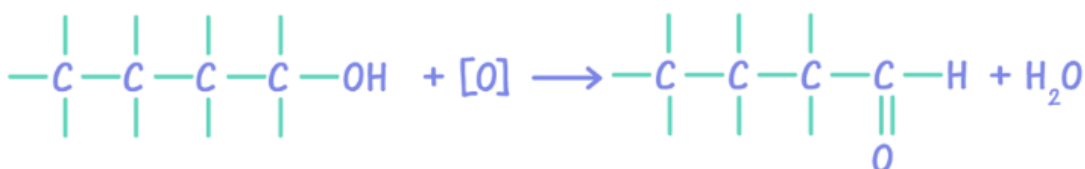


21.1.2 Aldehyde formation

Oxidation of primary alcohols



Example





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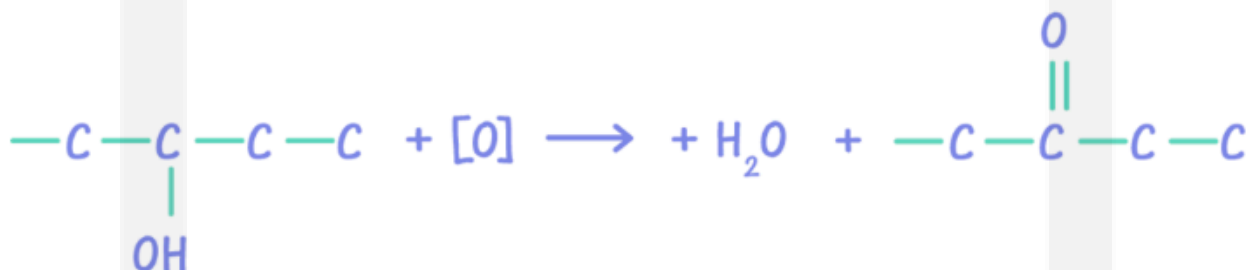
- acidified potassium dichromate $K_2Cr_2O_7$ or acidified $KMnO_4$, can be used as reagent, and distill off the product to produce aldehydes
- heat but no reflux, to make sure a carboxylic acid isn't formed instead

21.1.3 Ketone formation

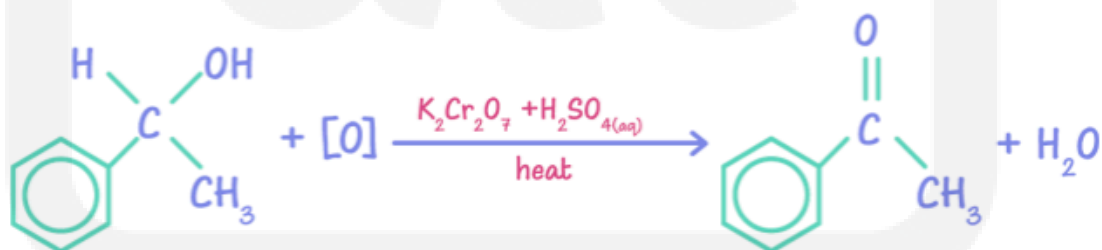
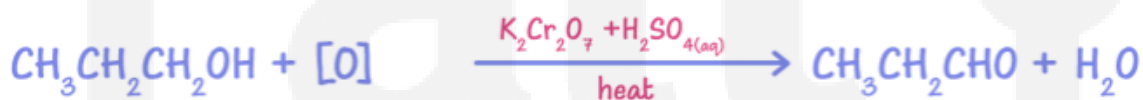
Oxidation of secondary alcohols



Example



- acidified potassium dichromate $K_2Cr_2O_7$ or acidified $KMnO_4$, used as reagent. and distill off the product to produce ketone
- heat, under reflux



21.2 Reactions of carbonyl compounds



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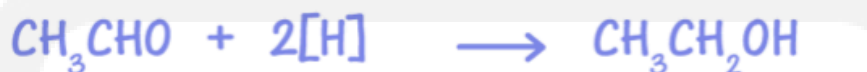
21 Carbonyl compounds

21.2.1 Reduction of carbonyls

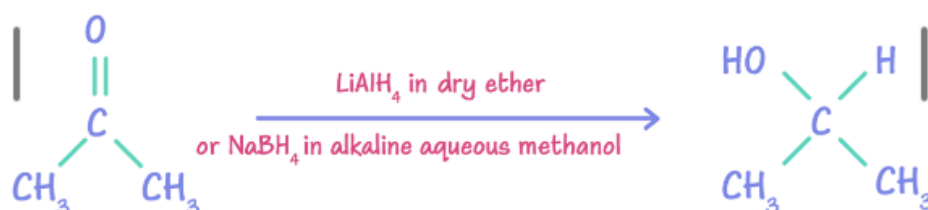
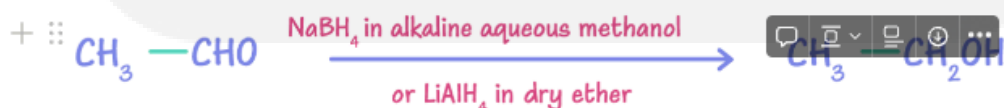
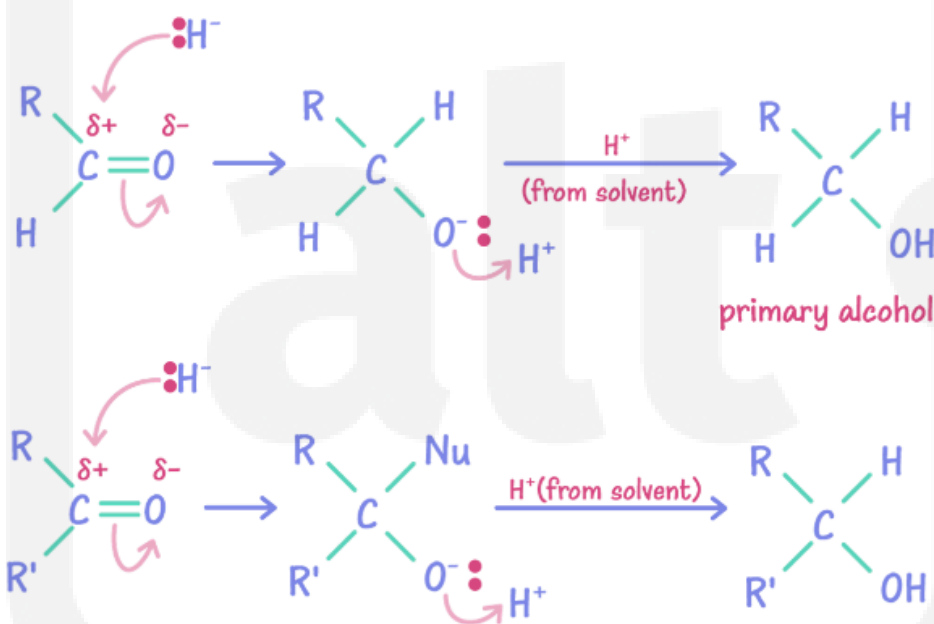
Reagents: sodium borohydride, NaBH_4 (aqueous or alcoholic solution) - weaker reducing agent

or lithium aluminium hydride, LiAlH_4 (in dry ether) - stronger reducing agent

Products: aldehydes $\rightarrow$ primary alcohols / ketones $\rightarrow$ secondary alcohols



hydrogen from
reducing agent





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Common name	sodium <u>borohydride</u>	lithium <u>aluminium</u> hydride
Systematic name	sodium <u>tetrahydridoborate(III)</u>	lithium tetrahydridoaluminate (<i>III</i>)
Formula	NaBH_4	LiAlH_4
Solvent used and reaction conditions	methanol-water mixture, room temperature or warm	totally dry ether, room temperature or at reflux (35°C)
How the product is separated	adding acid and extracting the product from the aqueous mixture with an organic solvent such as ether	adding water (with care), then acid, and separating the ether solution of the product from the aqueous layer
Hazards during use	relatively safe in alkaline solution, but evolves hydrogen gas with acids	can catch fire with water or when wet, and ether solvents form explosive mixtures with air
Functional groups that can be reduced, and what they are reduced to	carbonyl compounds (to alcohol)	carbonyl compounds (to alcohol) esters (to alcohols) carboxylic acids (to alcohols) amides (to amines) nitriles (to amines)

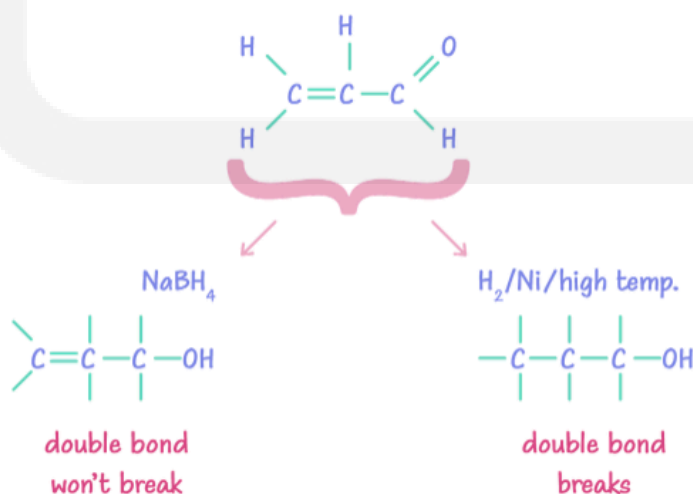
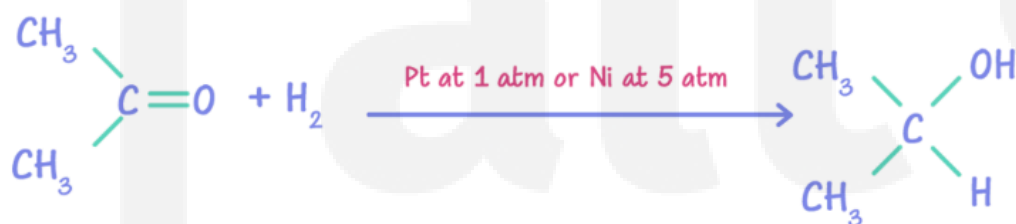
21.2.2 Reduction with Hydrogen

Reagent: hydrogen

Conditions: nickel or platinum catalyst / high temperature

Type of reactions: hydrogenation, reduction

Products: aldehydes $\rightarrow$ primary alcohols / ketones $\rightarrow$ secondary alcohols



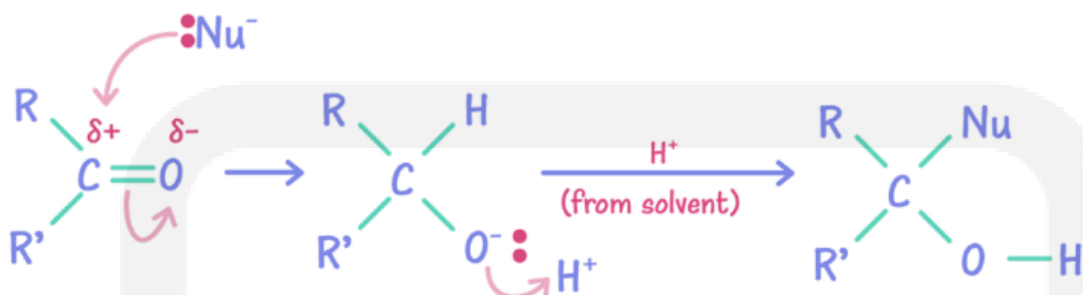


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21.2.3 Reaction with Hydrogen Cyanide (HCN)

Mechanism: nucleophilic addition



Reagent: HCN (in Na or KCN) (heat/reflux/alkaline solution)

→ HCN is *poisonous, not readily available*

→ HCN is *weak acid (proton donor)*, ionisation is little

→ *less dissociation of H^+ and CN^- , thus very slow reaction*

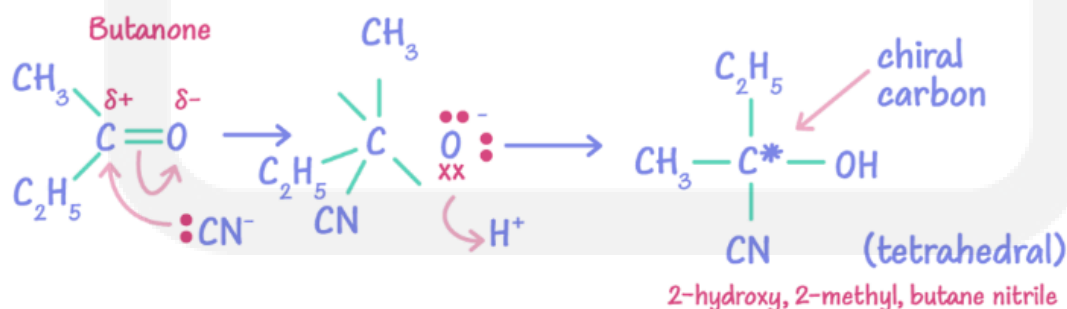


- Instead use NaCN catalyst and few drops of NaOH (to remove H^+), (aqueous, heat and reflux)



or make CN^- and H^+
'in situ (in reaction)'

Example



if the CN^- comes from above its positive } 50% +ve
if the CN^- comes from below its negative } 50% -ve } **racemic mixture**

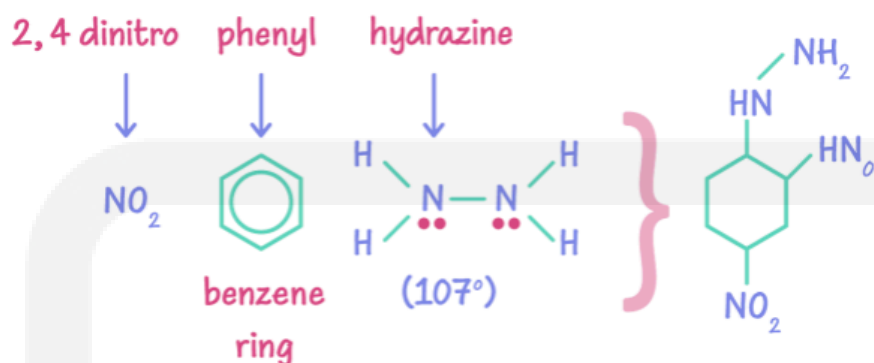
'A racemic mixture contains equal amounts of two enantiomers and has no effect on polarised light.'



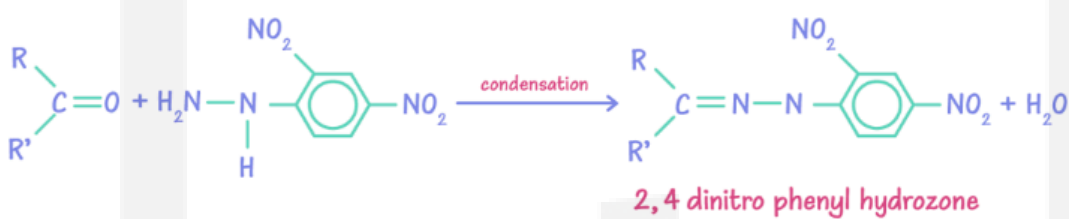
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21.3 Testing for presence of carbonyl compounds



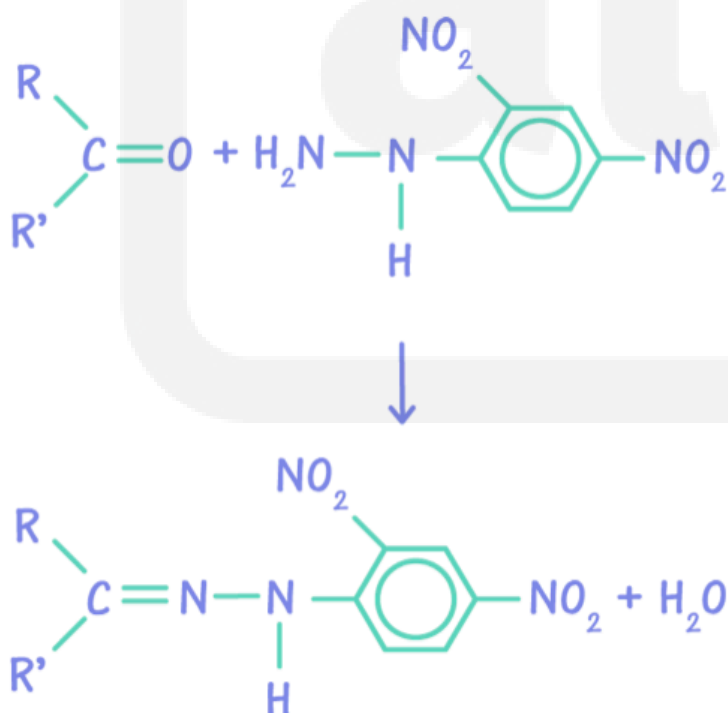
Aldehydes and ketones will form an orange precipitate with 2,4, dinitro phenyl hydrazine



Hydrazone: has sharp, well defined melting points

(product) can be used to identify the parent carbonyl compound with reference

General reaction:

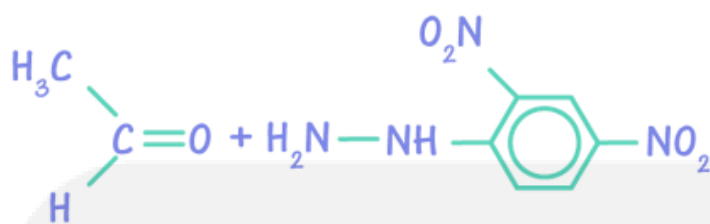




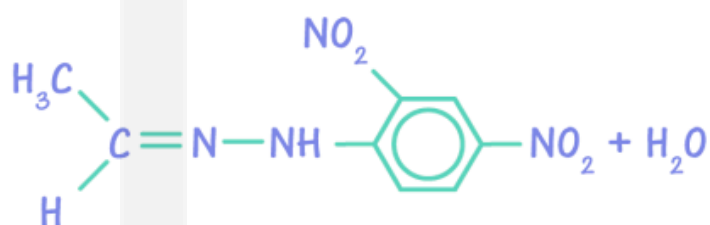
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Example:



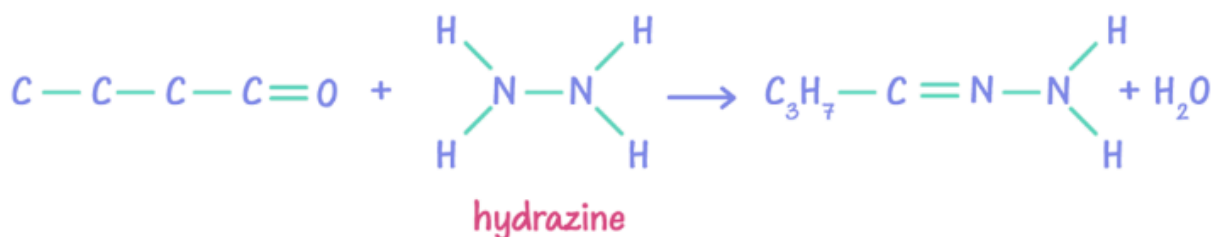
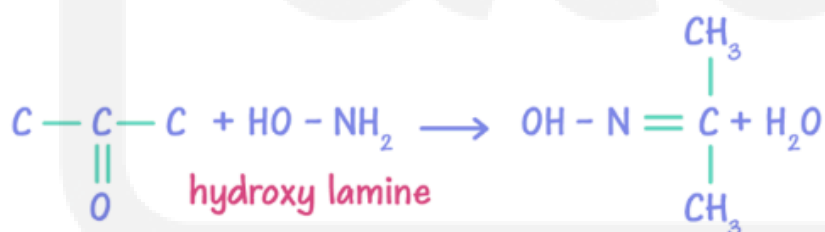
2, 4 dinitrophenylhydrazine



ethanal 2, 4 dinitrophenylhydrazone

Carbonyl compounds react in the same way with hydroxylamine and hydrazine (but there is no ppt)

Example





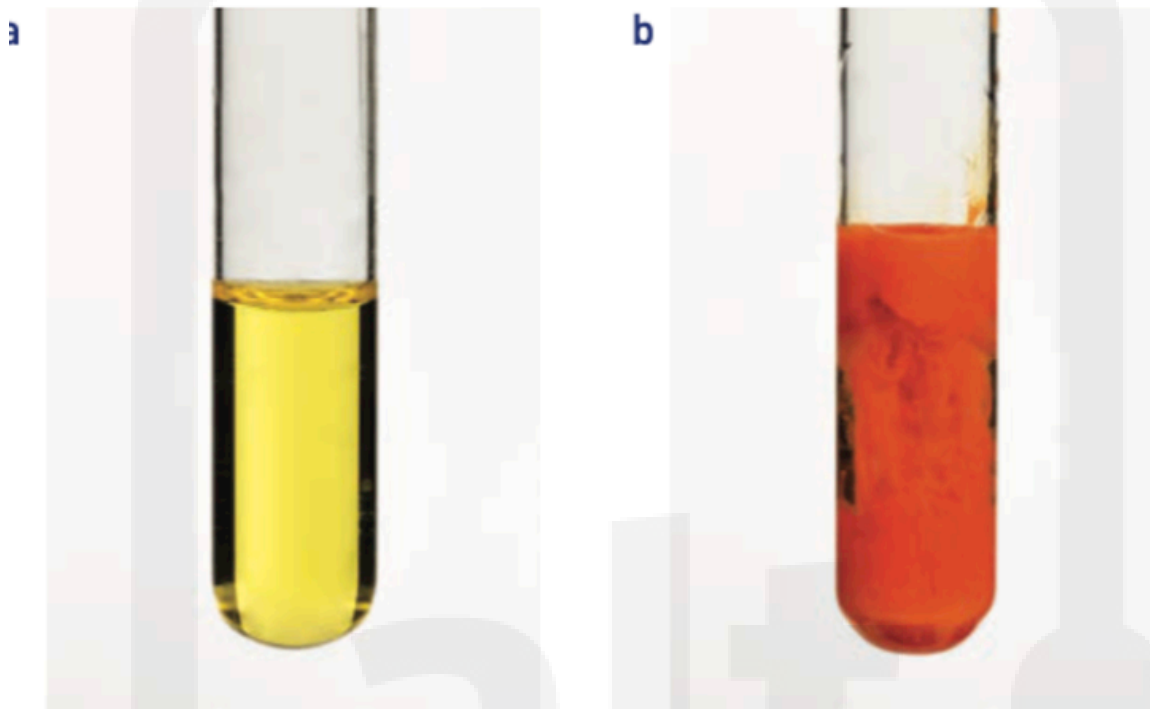
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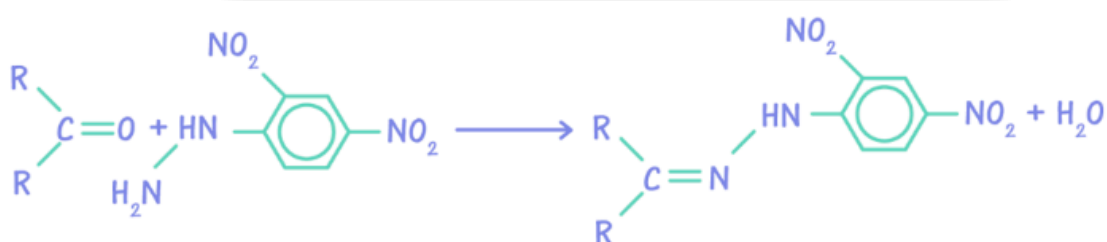
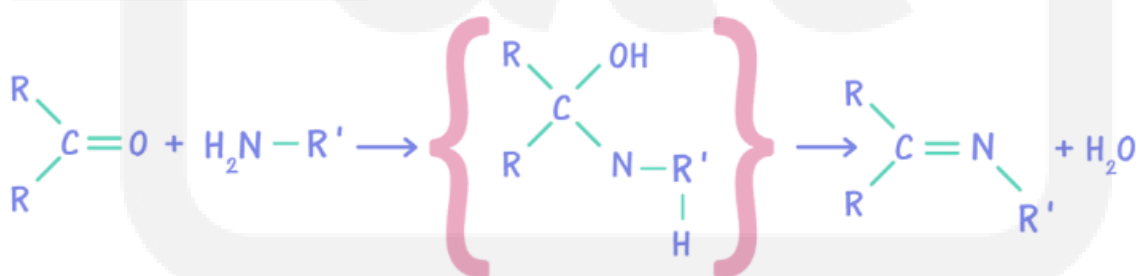
Conditions:

- 20°C room temperature
- no heat required because product may decompose

You can differentiate between aldehydes and ketones by using a mild oxidizing agent; aldehydes produce carboxylic acids, ketones remain ketones.



Condensation reactions:



2,4 DNPH

a 2,4-dinitrophenylhydrazone



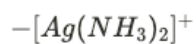
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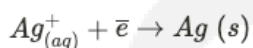
21.3 Testing for aldehydes and ketones

- **Tollens reagent:** ammonia and silver nitrate solution

☞ contains diamine silver(I) ion –



☞ the silver(I) ion is reduced to silver –



Observation: silver mirror



- **Fehling's solution:** contains a copper(II) complex ion giving a blue solution

☞ on warming, it will oxidise aliphatic (but not aromatic) aldehydes

☞ copper(II) is reduced to copper(I)

☞ red precipitate of copper(I) oxide, Cu_2O is formed



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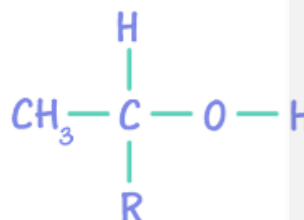
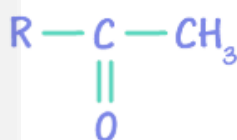


21.3.1 Iodoform reaction

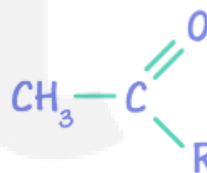
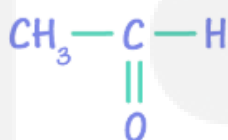
Tri-iodomethane test;

When warmed with a mixture of iodine and sodium hydroxide, methyl ketones (compounds containing the $\text{CH}_3\text{CO}-$ group) produces a yellow ppt of iodoform (triiodomethane).

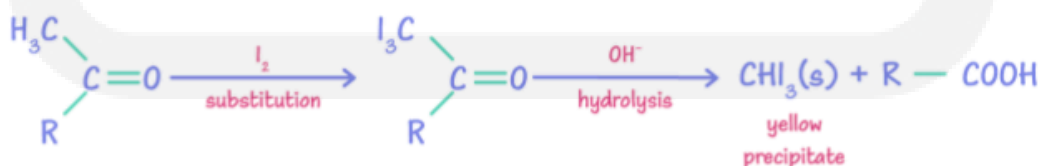
positive for methyl ketones



positive for ethanal (aldehyde)



General reaction



Overall equation





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Example



- Tri-iodomethane, CHI_3 , is a pale yellow insoluble solid, with a sweet, antiseptic smell.

❓ Skill Check 1

1. Compound Q

- contains a chiral center,
- gives a positive result with Fehling's reagent
- gives a positive result with alkaline aqueous iodine.

What could compound Q be?

- A 1-hydroxybutanone
- B 2-hydroxybutanal
- C 3-hydroxybutanal
- D 3-hydroxybutanone

▼ Solution

[2018 JUN P11 Q25]

C

For A, on the first carbon atom a hydroxy group must be present. Therefore, ketone can be on third or second carbon but there is no chiral carbon.

For B, carbon is attached to 4 different groups (chiral carbon), it is an aldehyde so gives a positive result with Fehling's reagent. There is no $\text{CH}_3\text{C}=\text{O}$ group, hence it doesn't give a positive result with alkaline aqueous iodine.

For C, hydroxy group is attached to the third carbon, there is a chiral carbon, it is an aldehyde and CH_3 group is directly attached to a carbon atom having an OH group so it also gives a positive result with alkaline aqueous iodine. Hence, C is the correct option.



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Points to Note

- If the product of carbonyl reduction has a chiral carbon, the product will be a racemic mixture.
- Glucose will also react with tollens reagent.
- A good test for the presence of a carbonyl compound is the formation of an orange precipitate when a solution of 2,4-DNPH is added to an unknown compound.
- Methanal (also known as formaldehyde) is also an aldehyde
- A mixture of potassium iodide and sodium chlorate(I) is an alternative reagent for the triiodomethane reaction.

alt

